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Research Program/Project/Study Title: Acceptability of Squash Pineapple Cake	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Roxas City	
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(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Agriculture/Fruit and Vegetables		(8) Discipline/Field of Specialization Food Development	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The purpose of the study is to have unique dessert from quash pineapple cake in order to maximized and utilize the use of our raw materials and elevate ones product into a more higher quality in order by the people to taste something new.

This experimental-developmental study was able to develop a squash vegetable into dessert with pineapple into sweet dessert. The statistical tools used in the analysis and interpretation of data were the Mean and the Analysis of Variance (ANOVA) set at 0.01 alpha level. Squash cake pineapple cake had shown to be liked extremely by 10 semi-trained panelist. on the other hand consumer preferences based on their acceptability of the three treatment of squash pineapple cake most list were in order (from highest to lowest) treatment C, A and B in terms of appearance, aroma, taste and texture.

Keywords: squash cake, pineapple

(10) Introduction

Food is one of the basic needs of human so we tend to cultivate a way on how to make it more interesting and more satisfying in such a way that saves money, time and effort. In addition, Human life doesn't have satisfaction so we are eager to find a solution on how to make our self-satisfy and sometimes the alternatives that we found out had a better result than the original one.

Squash is a vegetables consume by Filipino people which are abundant in the Philippines, a vegetables proven to be very useful as main ingredients or substitute component in food preparation. Squash is rich in vitamin A that good for the eyesight, resistance to infection, normal growth and healthy skin.

This study was conducted in order to maximized the use of our raw materials and elevate ones product into a more higher quality in order by the people to taste something new.

Therefore the conduct of making squash cake with pineapple syrup it serve as reference to those who are interested to venture in making variety of cake flavor for family consumption or for business. Hence the result of this research is a vital for future researcher who are interested in this field, using other ingredients in making cake of great quality that might other source of income.

Objectives: The general objective of this study aims to develop a nutritious cake from squash vegetables with pineapple . Specifically, it seeks;

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1. To determine the sensory qualities of squash pineapple cake in terms of appearance, aroma, texture, and taste among three treatments;
2. To find out the acceptability of squash pineapple cake syrup in terms of appearance, aroma, texture, and taste among three treatments;

Squash is a part of the same family as the pumpkin and zucchini. It is often thought of as a fruit due to the presence of seeds, and in some cases, people prefer to eat it raw. The squash was an imported crop during the pre-Columbian Era.

Squash was a trendy shape in the Moche ceramics from Northern Peru. It is one of the earliest plants domesticated in America. Five species of squash were domesticated independently before European contact. Wild squashes are extremely toxic to humans and other mammals; wild squash contains cucurbitaceous which are poisonous to smaller bodied mammals such as humans. It is a flowering plant which is usually served cooked, and its seeds and blossoms are also cooked and eaten. There are various types of squashes that are edible. Zucchini, crookneck, pattypan and such are quick-growing types. The winter squashes are large fruited; these fruits can be stored for many months after their season has passed. Evidence shows that the consumption of squash started around 75000 years ago in America and Mexico. The European settled transported squash seeds back to their land. Squash is thought to be a mix of pumpkin and zucchini. Here are 11 reasons why squash needs to be a part of your diet. Although originating in the Americas, squash are now grown in most countries. While squash tend to be quite nutritious, with high levels of vitamins A and C, niacin, riboflavin, and iron, their attraction to humans extends beyond this to more internal and aesthetic values. Their great variety in colors, color patterns, and shapes—from light green or white to deep yellow, orange, and dark green, from solid to striped, and from flattened to cylindrical to crookneck varieties—combined with their special aroma and taste, offers humans a unique visual and culinary experience.

Pineapples are tropical fruit that are rich in vitamins, enzymes and antioxidants. They may help boost the immune system, build strong bones and aid indigestion. Also, despite their sweetness, pineapples are low in calories. Pineapples are members of the bromeliad family, and one of the few bromeliads to produce edible fruit, according to the biology department at Union County College. The fruit is actually made of many individual berries that fuse together around a central core. Each pineapple scale is an individual berry.

Pineapples' nutritional benefits are as fascinating as their anatomy. "Pineapples contain high amounts of vitamin C and manganese," said San Diego-based nutritionist Laura Flores. These tropical treats are also a good way to get important dietary fiber and bromelain (an enzyme). "As well as having high amounts of manganese, which is important for antioxidant defenses, pineapples also contain high amounts of thiamin, a B vitamin that is involved in energy production," Flores said.

Synthesis:

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The product development studies include product formulation and utilization and commercialization

(11)

1. Product Formulation and Utilization

Procedure in the Preparation of Squash vegetables. Prepare the tools and equipment needed. Select good variety of squash and pineapple. Wash and pare the skin, shred. Set aside for later use

Procedure in the Preparation of squash pineapple cake.

1. Mise en place ingredients, utensils and equipment needed.
2. Pre heat the oven for 15 minutes.
3. Sift the four ingredients, all purpose flour, baking powder, baking soda, iodized salt.
4. In a separate bowl, combine oil and sugar. Add egg one at a time, beating well each after each addition.
5. Stir in the vanilla, add the dry ingredients, then add the grated squash and pineapple
6. Mix until well-blended. Pour into prepared round pan and bake for 35 minutes or until done
7. Squash cake pineapple product

The Nine (9) Point Hedonic rating scale used to determine the sensory qualities in terms of appearance, aroma, taste, and texture, and acceptability of the product.

The experiment was carried out in three treatments with the different amount of squash which is 50 grams treatment A, 75 grams treatment B, 100 grams treatment C. To obtain the desired result and products the process of experimental trial and error was done.

2. Production of technology manual/cooking guide/GMP manual

The quality standards, cooking guide, and other IEC materials will be developed and produced. It will be disseminated and distributed to various interested stakeholders.

The experimental design will be CRD with 3 treatments and 3 replicates for each grouping. The gathered data will be computed using the SPSS and analyzed using Analysis of Variance (ANOVA) at 0.01 percent level of significance.

(12) Results and Discussions

Sensory Evaluation of squash pineapple cake

Table 1 reflects the preferences of the 10 semi-trained panelist of squash pineapple cake in three treatments on the sensory quality in terms of appearance, aroma, taste and texture.

In terms of appearance, all treatments are extremely appealing, in terms of aroma, treatment C and B are extremely pleasant and treatment A is very much pleasant, in terms of taste

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all treatment are extremely delicious and in terms of texture treatment C got the mean score of 8.8 interpreted as extremely smooth.

Table 1. Sensory qualities of squash cake with pineapple syrup

Sensory Qualities	Treatment A		Treatment B		Treatment C	
	Mean	AD	Mean	AD	Mean	AD
Appearance	8.7	EA	8.5	EA	8.9	EA
Aroma	7.9	VMP	8.6	EP	8.9	EP
Taste	8.7	ED	9.1	ED	9.2	ED
Texture	8.0	VMS	8.1	VMS	8.8	ES

Legend:

Scale of Means
8.12 – 9.00

Qualitative Description
Extremely Appealing
Extremely Pleasant
Extremely Delicious
Extremely smooth

Table 2 reflect the consumer evaluation on the acceptability of squash cake with pineapple syrup shows that in terms of appearance treatment C got the mean score of 8.40 interpreted as liked extremely, in terms of aroma treatment C got the mean of 8.40, in terms of taste got the mean score 8.57 and in terms of texture treatment c got the mean score of 8.48, and for acceptability all treatment are acceptable by the consumer which interpreted as extremely acceptable.

Table 2. General Acceptability of Squash Pineapple Cake as evaluated by Consumers.

TREATMENTS	A		B		C	
Quality Attributes	Mean	AD	Mean	AD	Mean	AD
Appearance	8.23	LE	8.20	LE	8.40	LE
Aroma	8.33	LE	8.28	LE	8.40	LE
Taste	8.33	LE	8.32	LE	8.57	LE

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Texture	8.47	LE	8.43	LE	8.53	LE
Acceptability	8.34	LE	8.31	LE	8.48	LE

Legend:

Scale of Means

8.12 – 9.00

7.23 – 8.11

6.34 – 7.22

Qualitative Description

Liked Extremely

Liked Very Much

Liked Moderately

(13) Conclusions and Recommendations

Conclusions

Based on the findings of the study, the following conclusions are drawn:

Squash pineapple cake can be used the thee treatment in terms of sensory quality.

On the other hand, consumers' preferences based on the acceptability all treatments are acceptable.

Recommendations

All treatment are recommended in making Squash pineapple cake. .

I was also suggested to have an individual packaging of squash pineapple cake.

It is further recommended to housewives, consumers, home economics and food technologist to make and squash pine apple cake.

Further study may be conducted considering other variables not included in this study.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Squash cake with pineapple syrup can be produced and can be commercialize

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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